

Topography & Bathymetry Profiles, Depth-Age Relationship and Plate Motions

Note that this is an open book exercise, it is not a group exercise and to be completed by you individually

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****File name format for submission**** LastnameFirstname_ERS312Lab1

Submit all answers and graphs in one pdf document through Quercus (Assignments --> Assignment 1).

Part I – Bathymetry profiles and GeoMapApp

Objectives: (1) Plot topography and bathymetry (ocean depth) to relate large-scale features on Earth's surface to tectonic plates, earthquakes, and volcanoes. (2) This exercise also provides experience with graphing with GeoMapApp and concepts of scale.

A bathymetric profile (cross-section) provides a "skyline view" of the sea floor in which hills are seen as rises and valleys as depressions. For a graphical profile to illustrate the true shape of the sea floor, vertical and horizontal distances must be the same or have a ratio of 1:1. This means that one unit on the vertical scale is the same distance as one unit on the horizontal scale. However, the slopes of ocean features (i.e., relief) are generally so small that it is difficult to see sea floor features. Typical Atlantic Ocean basin features are only a few kilometers high, while the basin itself extends laterally for thousands of kilometers. If the profile is displayed with a 1:1 ratio on a regular sheet of paper, it would appear as a flat line.

In order to illustrate the details of the sea floor relief, the depth scale of an ocean profile is vertically exaggerated (stretched) relative to the horizontal scale. Vertical exaggeration causes distortion in the shapes of the bathymetric features that are being illustrated, with the amount of distortion increasing with the amount of exaggeration. As vertical exaggeration increases on a profile, hills appear to be higher, valleys deeper and the slopes between them become much steeper.

PROFILING EARTH'S SURFACE USING GEOMAPAPP

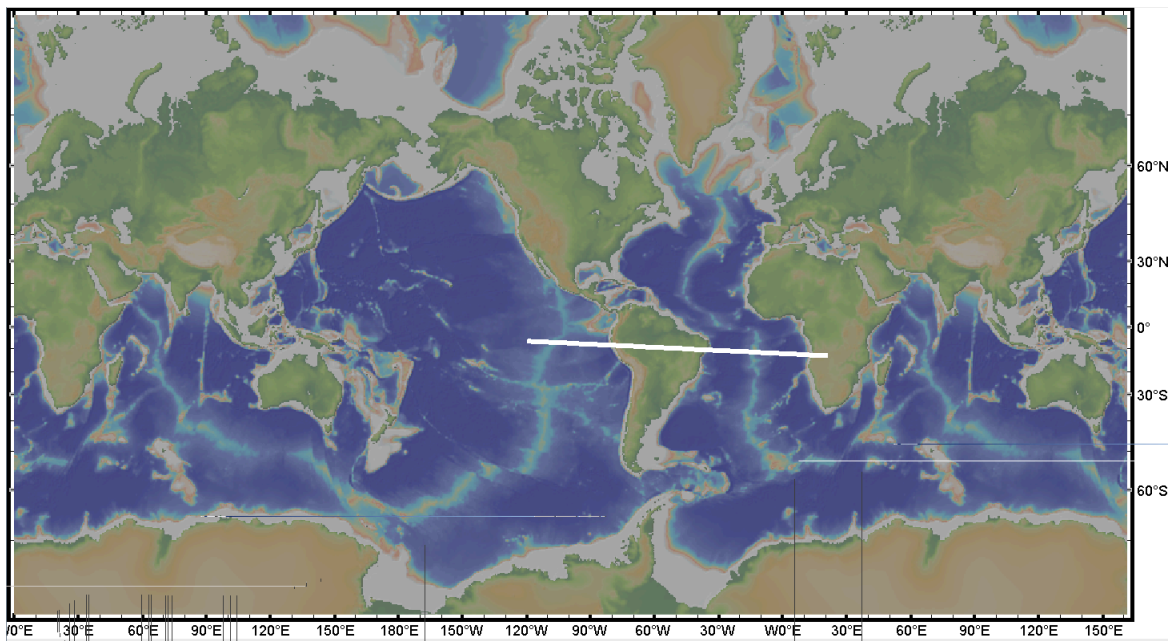
INSTRUCTIONS

Download GeoMapApp onto your computer or laptop. Go to <https://www.geomapapp.org/> and follow the instructions for downloading. After downloading the program, double-click on the GeoMapApp.jar icon to run the program.

Inside GeoMapApp, click on the Distance/Profile tool at the top of the screen:

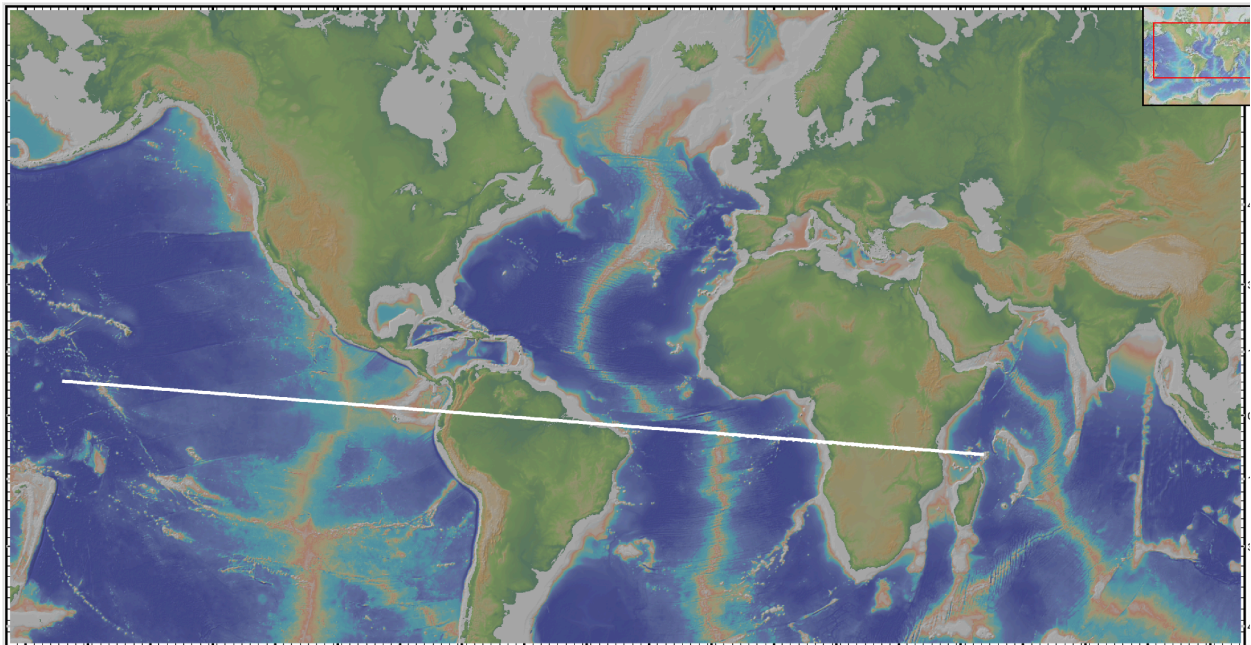


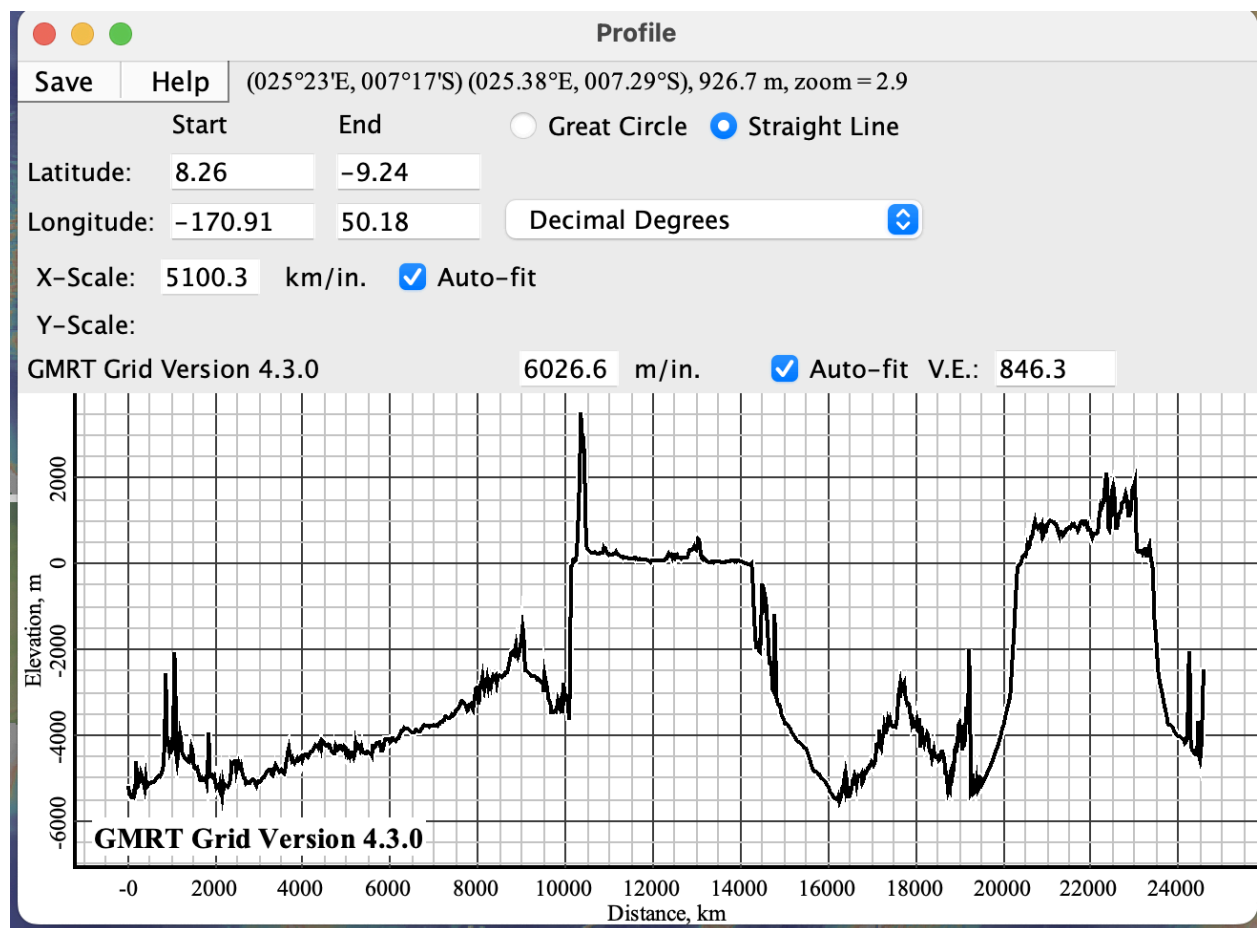
You may need to patiently wait while GeoMapApp opens the profiling tool. When the program is ready, refer to the world map below:



1. Pacific to Africa Profile: To draw the white line in GeoMapApp, place your cursor in the appropriate location on the Pacific Plate (B). Click on the mouse and hold the button as you drag the cursor over just north of Madagascar (B'). When you first do this, the line might be drawn as a curve (great circle). In the profile tool, click the box specifying that you would like to draw the cross section along a **straight line**:

Next, take a screenshot and **Paste your profile here** :





Add the following characteristics specified in Q1.1-1.3 to your profile. For some of these, you will need to refer to outside sources such as your notes.

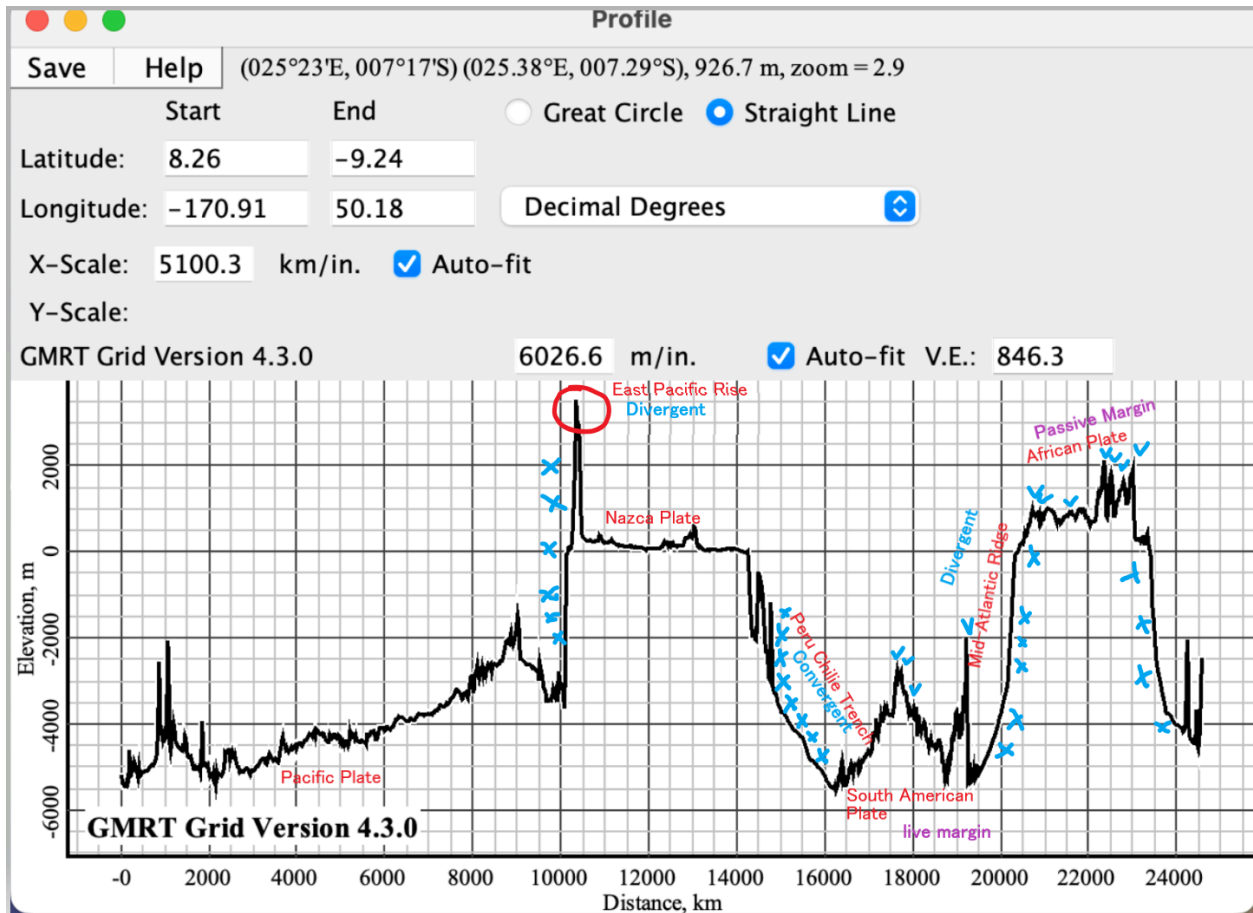
Profile 1 Questions:

★Q1.1 (9 pts): Identify and label the positions of the following features on the above profile (put labels on screenshot):

- a. Plate boundaries and their types (e.g., label East Pacific, Mid-Atlantic Ridge, show location of Peru-Chile trench)
- b. Tectonic plates and their names
- c. Continental margins and their types (i.e., passive or active)

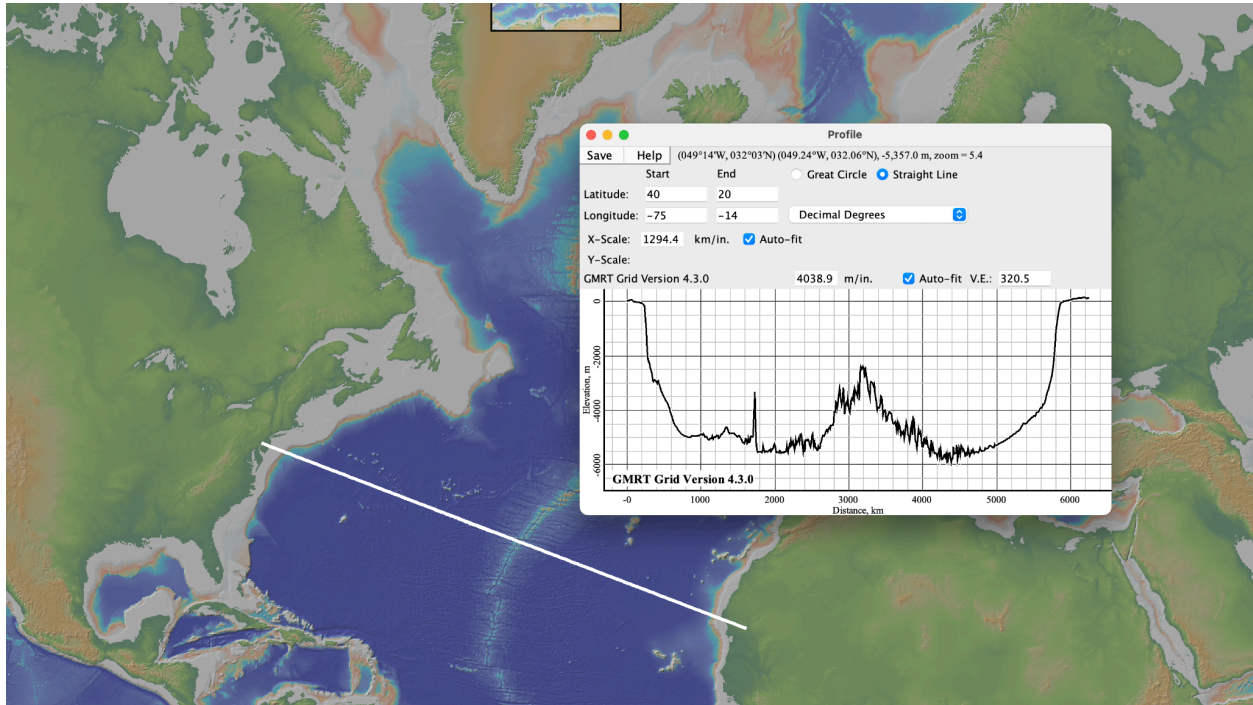
★**Q1.2 (3pts):** Put a series of X's within the profile to represent zones where earthquake activity might be expected to occur.

★**Q1.3 (3pts):** Put V's on the surface of the profile to illustrate where you would expect volcanic activity to occur.



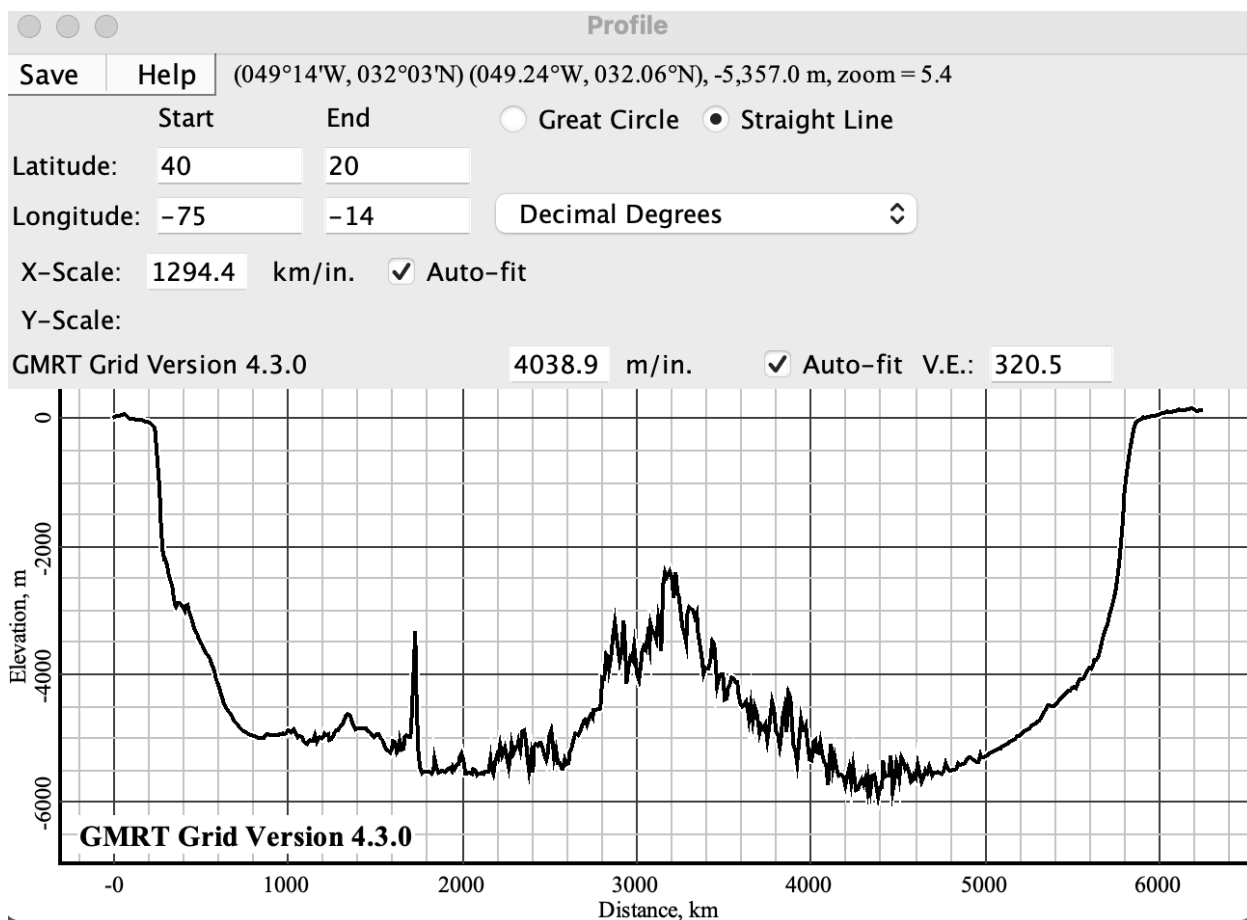
2. Atlantic Ocean Basin

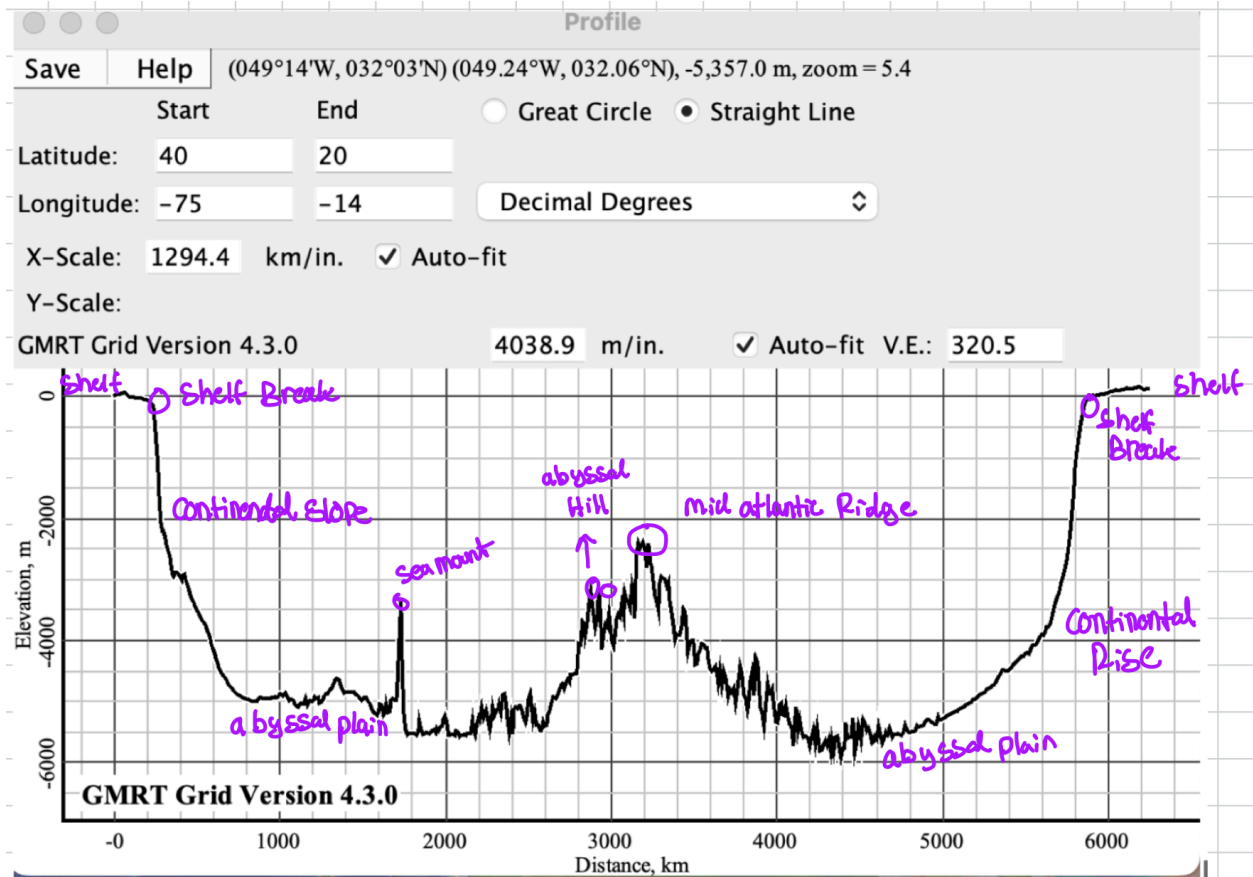
Make a bathymetric profile across the Atlantic from N. America (~75°W, 40°N) to N. Africa (~14°W, 20°N). Try to include only a little bit of the land above sea level. Take a screenshot and **Paste your profile here:**



Profile 2 Questions:

★Q2.1 (7pts): On the above profile, label the following features of the ocean basin: **shelf break, shelf, continental slope, abyssal plain, mid-Atlantic ridge, continental rise, seamount OR abyssal hill.**





★Q2.2 (4pts): What is the approximate ocean **depth** at the mid-Atlantic ridge and the **relief** of the ridge (i.e., difference in depths, or elevations, between the top of the ridge and the adjacent, relatively flat ocean bottom)?

Ocean depth at mid-Atlantic ridge : ~ **2500 meters**

Relief (difference in ocean depth between the ridge and the abyssal plain) of the mid-Atlantic ridge : $6000 - 2500 = \sim \mathbf{3500 \text{ meters}}$

★Q2.3 (3pts): What is the approximate **slope** of the **west flank** of the mid-Atlantic Ridge (i.e., measure the difference in ocean depth along the profile between 2200 km and 3200 km distance and divide by the difference in distance, $3200 - 2200$ or 1000 km). Be sure that both measurements, differences in depth, and distance, are in the **same units**, either km or m. The resulting number will be the slope expressed as a ratio (show calculation).

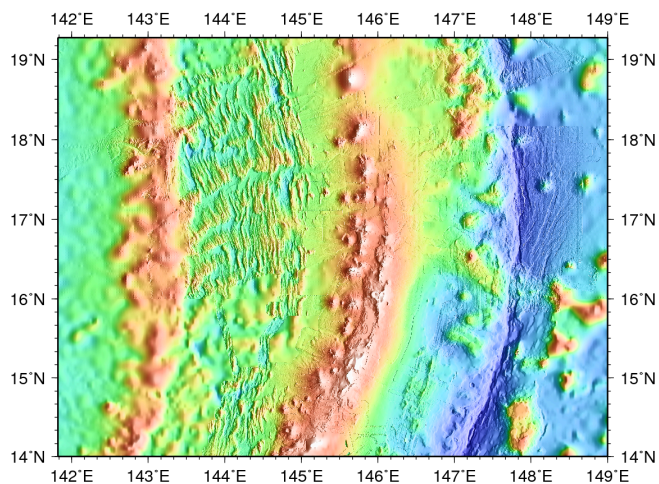
$\Delta\text{depth} = \text{Depth at 3200 km} - \text{Depth at 2200 km}$

$$= -5000 + 2500 = -2500 \text{ m}$$

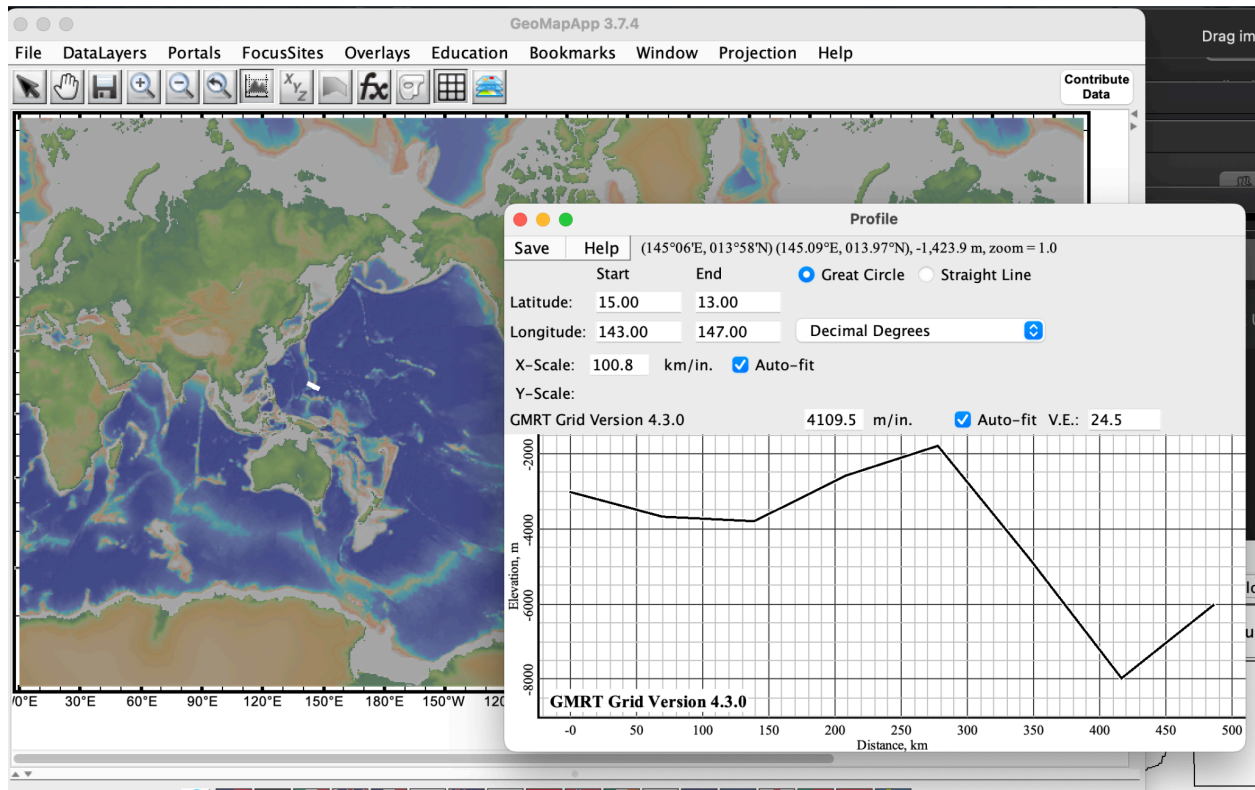
$$\text{slope} = \Delta\text{depth} / \Delta\text{distance} = -2500 / 1000 = -2.5 \text{ m/km}$$

Approximate slope of the west side of the mid-Atlantic ridge: -2.5 m/km

3. Mariana Trench region

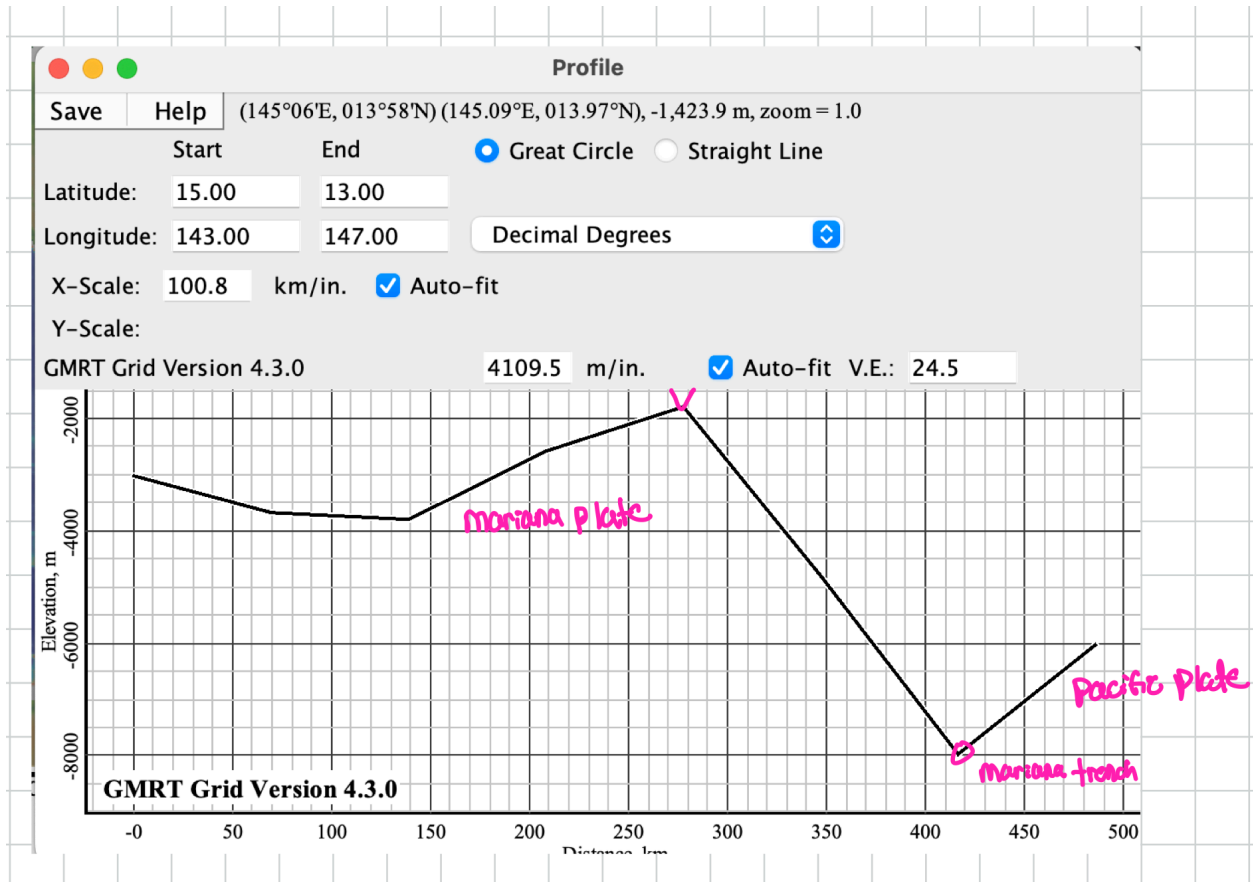


Produce a bathymetric profile across the Mariana Trench region (draw a line 15°N , 143°E to 13°N , 147°E). Take a screenshot of the profile and **Paste your profile here:**



Profile 3 Questions:

★Q3.1 (4pts): On the above bathymetric profile, label the trench, the Mariana Plate, the Pacific Plate and put a V on the surface of the profile to illustrate where you would expect volcanic activity to occur.

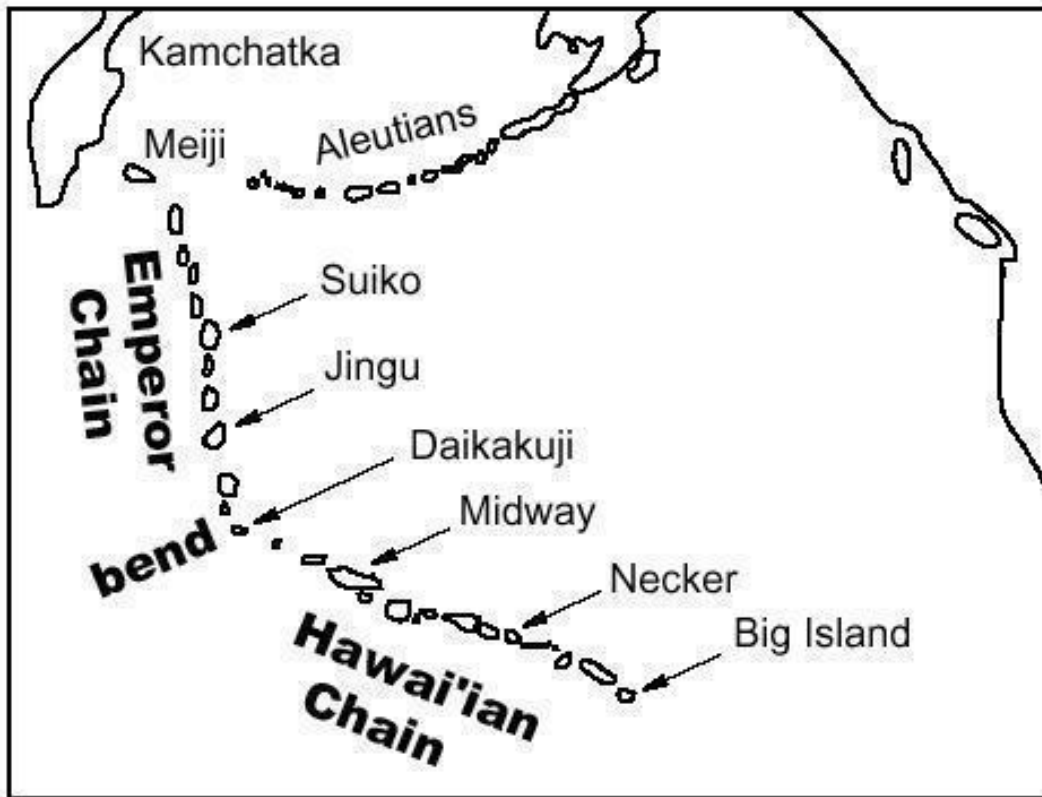


★Q3.2 (1pts): What do you think is happening in the crust and upper mantle along the above profile (i.e., what are the plates doing here?).

The Pacific Plate is subducting beneath the Mariana Plate at the Mariana Trench, forming a deep subduction zone.

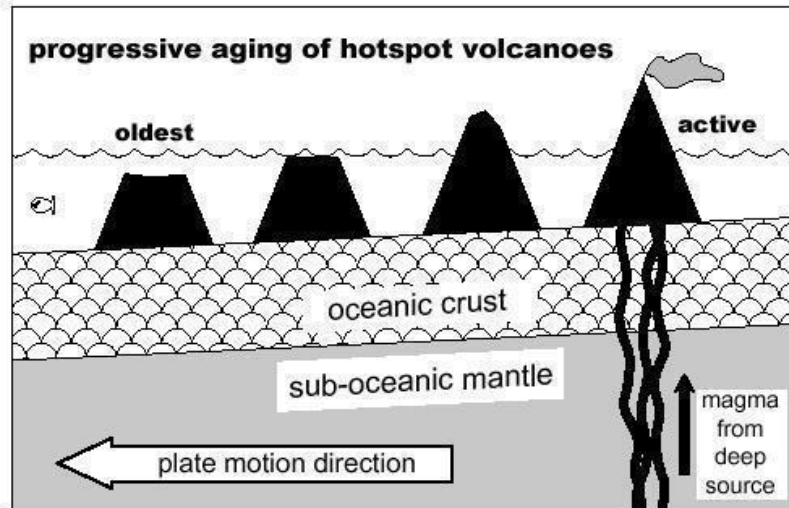
4. Hawai'i-Emperor chain

The Hawai'i-Emperor chain of seamounts (a more correct term would be tablemounts) stretches from its active end at the Big Island of Hawai'i west and north across the Pacific Ocean floor to the Aleutian trench near the Kamchatka Peninsula (first figure). There are about 110 individual volcanoes in the Hawai'i-Emperor chain (see the data table), which is about 6000 km (3800 miles) long altogether. The Hawai'i-Emperor chain is divided into two segments, the WNW-trending Hawai'ian chain and the N-trending Emperor chain. The two chains meet at a prominent bend, around the underwater seamount Daikakuji.



The active end (youngest end) of the Hawai'i-Emperor chain is at the Big Island of Hawai'i with the Kilauea volcano. As one progresses towards the west-northwest, the volcanoes of the Hawai'ian Islands get progressively older (see data table). Once active volcano building through eruptions of lava ceases, the erosional forces of tropical weathering, landslides, river erosion, and wave action overcome the island, and erode it down to sea level (second figure). The extinct volcano evolves to a flat-topped mesa ringed by coral reefs, and then to an atoll with nothing but the circular reef showing. Finally, the volcano sinks beneath the waves, and becomes an underwater tablemount.

The Hawai'i-Emperor chain is a classic example of a hot spot track. The standard explanation begins with a hot spot whose source of magma is rooted deep in the Earth's mantle. The hot spot magma source is thought to be fixed in the deeper mantle, with a slab of ocean crust and uppermost mantle (called a plate) moving laterally above the hot spot. As the Pacific Plate moves over the Hawai'ian hot spot, magma "punches up" through the Pacific Plate, creating an active volcano. Plate motion carries the active volcano away from the magma source, the volcano goes extinct, and a new volcano grows over the hot spot. As the extinct volcano is carried farther from the hot spot source, the volcano sinks beneath the waves mostly due to aging and cooling of the ocean crust underneath the extinct volcano; this cooling causes subsidence of the ocean floor.



Hot spot tracks are very important features for determining both the direction and speed of the plate upon which the seamounts rest. The direction of plate motion is given by the orientation of the chain of seamounts and volcanoes. Using your "hands of science", you can quickly determine that the plate "moves towards the oldest volcano." As can be seen in the first figure, the Pacific Plate moved almost due north during "Emperor time" (from 75 to 42 million years ago), and then changed direction about 42 million years ago (the age of the volcanoes at the bend in the chain), to move west-northwest during "Hawai'i time" (from 42 Ma to the present).

Hot spot tracks also give the speed of plate motion if the length of the chains of volcanoes and seamounts and the ages of the volcanoes and seamounts are known (data table). Plates typically move about 1–10 cm/year, which is equivalent to 10–100 km/Ma (kilometers per million years). These speeds are about the rates at which fingernails grow, and may seem rather slow on the human time scale, but are very fast on the geological time scale.

Questions:

★**Q4.1 (3pts):** Based on the age and distance relationship of the entire Hawai'i-Emperor chain given in the below table, what is the "typical" rate of Pacific Plate motion over the last 65 Ma (in cm/year, show calculation and units).

Solution :

① distance = 6000 km = 600,000,000 cm [converted to match desired sol. units]

Time = 65 million yrs = 65,000,000 yrs

$$\text{Rate} = \frac{600,000,000}{65,000,000} = \frac{600}{65} = \sim 9.23 \text{ cm/yr}$$

thus roughly the Rate is 9.24 cm/yr.

Typical Rate (3pts): _____ 9.24 cm/yr _____

★Q4.2 (12pts): For the second part of the exercise you need to determine if the Pacific Plate sped up or slowed down over the last 65 Ma. In order to find that out you need to calculate, based on the below table, the rate of plate motion for both the Hawai'i Time and the Emperor Time separately (two calculations).

a. What is the rate of plate motion (cm/year, show calculation and units) for the

Emperor 'Time'? _____ (3 points)

Emporer Time :

Drakakusi

age = 42.4 ma

dist: 3493 km

.....

Mesji

.....

dist: 6000 km

age: 7 ma

$$\Delta \text{age: } 75 - 42.4 = 32.6 \text{ ma}$$

$$\Delta \text{dist: } 6000 - 3493 = 2507 \text{ km}$$

$$\text{Rate} = \frac{2507 \text{ km}}{22.6 \text{ ma}} \sim 7.64 \text{ cm/yr}$$

Emporer Time :

Drakakusi

age = 42.4 ma

dist: 3493 km

.....

Mesji

.....

dist: 6000 km

age: 6 ma

.....

$$\Delta \text{age: } 65 - 42.4 = 22.6 \text{ ma}$$

$$\Delta \text{dist: } 6000 - 3493 = 2507 \text{ km}$$

$$\text{Rate} = \frac{2507 \text{ km}}{22.6 \text{ ma}} \sim 11.09 \text{ cm/yr}$$

Hawai'an 'Time'? _____ (3 points)

4.2

Hawaiian Time

a.)

Daiikakujii:

time: 42.4 Ma

dist: 3493 km



distcm = 349300000

.....

$\Delta \text{distcm} = 349300000$

$\Delta \text{Time} = 42.4 - 0.2 = 42.2 \text{ Ma} = 42,200,000 \text{ yrs}$

$$\text{Rate} = \frac{349300000}{42,200,000} = \sim 8.28 \text{ cm/yr}$$

Kilauea

time: 0.2 Ma

dist: 0 km



distcm = 0cm

b. Has the Pacific Plate sped up or slowed down over the last 65 Ma (2pts)?

Since the plate motion has increased over time, we can say that the Pacific Plate has sped up over the last 65 Ma. (based on calc)

c. Approximately what direction did the Pacific Plate move before 42 million years ago (2pts)? Northward After 42 million years ago (2pts)? West-Northwest

volcano	age	age uncertainty	distance	distance uncertainty
name	(Ma)	($\pm$ Ma)	(km)	($\pm$ km)
Kilauea	0.2	0.2	0	1.5
Mauna Kea	0.38	0.05	54	1.8
Kohala	0.43	0.02	100	2
East Maui	0.75	0.04	182	2.5
Kahoolawe	1.03	0.18	185	2.5
West Maui	1.32	0.04	221	2.7
Lanai	1.28	0.04	226	2.7
East Molokai	1.76	0.07	256	2.9
West Molokai	1.9	0.06	280	3
Koolau	2.6	0.1	339	3.3
Waianae	3.7	0.1	374	3.5
Kauai	5.1	0.2	519	4.2
Niihau	4.89	0.11	565	4.5
Nihoa	7.2	0.3	780	5.6
unnamed 1	9.6	0.8	913	6.3
Necker	10.3	0.4	1058	7.1
La Perouse	12	0.4	1209	7.9
Brooks Bank	13	0.6	1256	8.2
Gardner	12.3	1	1435	9.1
Laysan	19.9	0.3	1818	11.1
Northampton	26.6	2.7	1841	11.3
Pearl & Hermes	20.6	0.5	2291	13.6
Midway	27.7	0.6	2432	14.4
unnamed 2	28	0.4	2600	15.3
unnamed 3	27.4	0.5	2825	16.5
Colahan	38.6	0.3	3128	18.1
Abbott	38.7	0.9	3280	18.9
Daikakuji	42.4	2.3	3493	20
Yuryaku	43.4	1.6	3520	20.1
Kimmei	39.9	1.2	3668	20.9
Koko	48.1	0.8	3758	21.4
Ojin	55.2	0.7	4102	23.2
Jingu	55.4	0.9	4175	23.6
Nintoku	56.2	0.6	4452	25.1
Suiko 1	59.6	0.6	4794	26.9
Suiko 2	64.7	1.1	4860	27.2

